



VCS VERIFICATION REPORT

HEILONGJIANG BIANFUSHAN WIND
POWER PROJECT

(VCS PROJECT ID: 1049)



Document Prepared By

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Summary:

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions reported for the project activity "Heilongjiang Bianfushan Wind Power Project" (VCS Project ID:1049) for the monitoring period 01/01/2013-31/12/2018 within the 1st VCS crediting period 25/12/2009-24/12/2019, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.3.

The verification was performed on the basis of VCS Standard Version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified one CL, no CAR nor FAR was raised in this monitoring period.

In CTI's opinion, the GHG emission reductions reported for the project in the monitoring report (version 02 dated 25/07/2022) are fairly stated. The GHG emission reductions of the period from 01/01/2013-31/12/2018 within the 1st VCS crediting period were calculated correctly on the basis of approved methodology ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources" (version 12.1.0), the monitoring plan contained in the registered VCS PD (i.e. registered CDM PDD version 03 dated 01/11/2011) for the 1st VCS crediting period 25/12/2009-24/12/2019.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

Hence, CTI is able to certify that the emission reductions from the “Heilongjiang Bianfushan Wind Power Project” during this monitoring period from 01/01/2013-31/12/2018 amount to 627,605 tCO₂e.

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1 INTRODUCTION

Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd. has commissioned Shenzhen CTI International Certification Co., Ltd (CTI) to carry out the verification and certification of emission reductions reported for the “Heilongjiang Bianfushan Wind Power Project” (the Project) for the monitoring period 01/01/2013-31/12/2018. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Heilongjiang Bianfushan Wind Power Project” for the period 01/01/2013-31/12/2018.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.2)/43/
- VCS Standard (version 4.3)/42/ and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 12.1.0)/46/ applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.3, the project size categorization belongs to 'Project' as the expected annual emission reductions is 125,030 tCO₂e lower than 300,000 tCO₂e. Therefore, according to the requirement of para 4.1.8 (4) of VCS standard Version 4.3 "The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.", CTI applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.2)/43/ and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: wind power project
- The project is not a grouped project.

Project Background

The project is located in Fengli Town, Jixian County, Shuangyashan City, Heilongjiang Province, and P.R.China. The site center location's geographical coordinates are east longitude of 130°51'28'' and north latitude of 46°44'35'' as verified by using GPS locator and checking map and project's layout/5/. The project activity was registered as a CDM project with Ref.5491 on 19/12/2011 with the crediting period starting from 19/12/2011. The project has been registered as VCS project with Ref. VCS 1049 under VCS Standard Version 3.3 with the start date 25/12/2009 when the commencement of operations activities started as confirmed by checking the operation log/8/, the VCS monitoring report and verification report for the period of 25/12/2009-18/12/2011 /32//33/.

The amount of 218,614 tCO₂e VCUs have been issued for the monitoring period 25/12/2009-18/12/2011 within the 1st VCS crediting period. GHG emission reductions have been applied for CDM for CERs within the 1st CDM crediting period, total 102,399 tCO₂e CERs have been issued for the monitoring periods 19/12/2011-31/12/2012.

The installed capacity of the project is 49.5MW equipped with 33 sets of wind turbine-generators in type of GE77-1500 with unit capacity of 1500kW, manufactured by General Electric (Shenyang)

Co., Ltd.. The electricity generated by the project activity is supplied to Northeast China Power Grid (NECPG), and the project is estimated to deliver 125,030 tCO₂e emission reduction annually.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered VCS PD (i.e. registered CDM PDD version 03 dated 01/11/2011) and monitoring plan contained in the registered VCS PD, the applied methodology, and the VCS Standard (version 4.3) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CTI appointed a verification team in accordance with CTI's internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader	Zhang Lei	1.1,1.2, 4.1,13.1	☒ DR ☒ SV ☒ RP ☐ TR
Team Member	Zhang Wenting	1.2	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report/1/, the registered VCS PD /16/, emission reduction calculation spreadsheet/2/, the following documents also were assessed as a part of the verification audit:

- The VCS validation report /32/;
- The previous VER/VCU monitoring report/32/ and verification report/33/, and previous CDM/CER monitoring report /35/and verification report/36/;
- Baseline and monitoring methodology ACM0002 (version 12.1.0) applied by the project/46/;
- Relevant decisions, clarifications and guidance from the Verra/42/-/45/; and
- Other information and references relevant to the project activity.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and
- An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 21/07/2022, CTI visited the project proponent Shuangyashan Longyuan Wind Power Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team /46/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /47/ are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Wang Zhewei	Station Manager	Shuangyashan Longyuan Wind Power Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity. Project's sustainable development contributions
Mr. Zhao Guochao	Operator	Shuangyashan Longyuan Wind Power Co., Ltd.	
Mr. Fu Tao	Operator	Shuangyashan Longyuan Wind Power Co., Ltd.	
Ms. Yang Huifang	Local resident	Local Village	The impact of the project activity; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity; Project's sustainable development contributions.
Mr. Li Jianhong			
Mr. Zhao Wei	Staff	Local Environment Protection Bureau	
Mr. Ma Qian	Project Manager	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.	Data collection and ER calculation.

2.4 Site Inspections

During the on-site assessment, CTI has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered VCS PD of the project activity;
- A review of information flows for generating, aggregating and reporting the monitoring parameters; and
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity sales receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 12.1.0), and the management system were assessed during the site visit.

2.5 Resolution of Findings

A corrective action request (CAR) shall be raised, where:

- i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- iii. Issues identified in a FAR during validation to be verified during verification have not been resolved by the project proponents.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

The verification team identified 1 CL in this monitoring period. No CAR was raised. The CL was satisfactorily addressed by the project proponents in the updated monitoring report version 02 dated 25/07/2022 (refer to Appendix D for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CTI confirmed that there was no FAR identified in previous verification/33/, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by Bureau Veritas Certification Holding SAS (BVCH) under the CDM rules and registered as CDM project on 19/12/2011, and then was validated by DNV Climate Change Services AS under VCS standard version 3.3.

CTI was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the project. By checking the UNFCCC website, the VCS pipeline and the previous verification reports/33/, it is confirmed the verification for this monitoring period 01/01/2013-31/12/2018 which is no more than consecutive 6 years is the first verification undertaken by CTI. Therefore, the verification undertaken by CTI is in line with the requirement for VVB stipulated in Section 4.1.20 of VCS standard version 4.3.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a CDM project on 19/12/2011 under the UNFCCC with Ref. 5491 with the crediting period starting from 19/12/2011. By checking the UNFCCC website (<https://cdm.unfccc.int/Projects/DB/BVQI1323445785.17/view>), it is confirmed that total 102,399tCO₂e of CERs have been issued for the monitoring periods 19/12/2011-31/12/2012 within the 1st CDM crediting period.

3.2 Methodology Deviations

The validation process /21/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 12.1.0) /46/. There was not any methodology deviation applied to this project. Details refer to section 4.1.

3.3 Project Description Deviations

By checking the registered VCS PD, the previous VCS monitoring report with the monitoring period 25/12/2009-18/12/2011, the previous CDM verification report with the monitoring period 19/12/2011-31/12/2012, documentation of the project's confirmation of the commissioning and start of operation of the wind turbine, Project construction-start application and permission/9/, Verification report of project construction completion/10/, and through the site visit interview with project owner, it is confirmed that the first turbine started commercial operation on 25/12/2009 which is the project start date according to the requirement of VCS. A deviation was identified for the crediting period of the Project. The project is registered as VCS project under VCS version 3.3 and completed validation before 19/03/2020. As per VCS requirement, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice. Therefore, the first renewable crediting period of the Project under VCS is updated from 25/12/2009-18/12/2011 to 25/12/2009-24/12/2019, which lasts for 10 years.

As per VCS requirement, the validation of crediting period renewal was not conducted within the two years after the end of the first VCS crediting period. Thus, the crediting period of the project ends on 24/12/2019.

CTI confirms that the project is fully implemented in accordance with the project description, and no any project description deviations. Details refer to section 4.1.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the “Heilongjiang Bianfushan Wind Power Project” for the monitoring period 01/01/2013-31/12/2018.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a wind power grid-connected plant and locates Fengli Town, Jixian County, Shuangyashan City, Heilongjiang Province, and P.R.China. The central geographical coordinates are verified on site by GPS system as east longitude of 130°51'28" and north latitude of 46°44'35". The total installed capacity is 49.5MW equipped with 33 sets of wind turbine-generators in type of GE77-1500 with unit capacity of 1500kW. It was confirmed by checking the operation logs/8/, the validation report, the previous VCS/CDM monitoring reports, verification reports /32/-/36/, Application Report for Commencement of Work/9/, Project construction-start application and permission/10/ and Verification report of project construction completion/11/, the project began to construct on 04/06/2009, and started commissioning on 25/12/2009. The actual implementation of the project during this verification period was verified in terms of nameplates of wind turbines /37/. The details of the turbines with respect to their installation and capacity have been verified to be consistent with description indicated in the registered VCS PD.

The electricity generated by the project activity is supplied to NECPG, which can be confirmed by the Power purchase agreement (PPA) signed between Shuangyashan Longyuan Wind Power Co., Ltd. and State Grid Heilongjiang Electric Power Ltd. /3/. All the monitoring system in operation period is consistent with the description in monitoring plan contained in the registered VCS PD. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the Operation logs/8/ and via on-site visit, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place/7/ and their implementation was confirmed by interview with the key operators/47/ and observing the operation.

CTI confirms that the project implementation is in accordance with the project description contained in registered VCS PD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

CL 01 was raised for lacking the manufacturer of the turbine-generators, which has been successfully closed, Refer to Appendix D for details.

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the monitoring plan contained in the VCS PD is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 12.1.0).

Compliance of monitoring with the monitoring plan

The monitoring has been carried out in accordance with the monitoring plan contained in the VCS PD/16/, CTI confirms that all parameters stated in the monitoring plan are monitored and reported appropriately. All parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (version 12.1.0) and the management system were assessed during the site visit. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

Parameters fixed ex ante

“Data and parameters fixed ex ante” in the MR are checked against the registered VCS PD. The parameter $EF_{grid,CM,y}$ is the combined margin emissions factor of the grid, which was determined ex-ante at the validation stage and has been fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the VCS PD.

Parameters monitored

CTI conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and electricity sales receipts;
- An identification that quality control and quality assurance procedures are in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the monitoring plan, the parameters to be monitored is as below:

$EG_{\text{facility}, y}$	Quantity of net electricity generation supplied by the project to the grid in year y
$EG_{\text{export}, y}$	Electricity exported to the grid by the project
$EG_{\text{import}, y}$	Electricity utilized by the project

Monitored parameters $EG_{\text{facility}, y}$, $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$

By site inspection and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CTI confirmed that the parameter $EG_{\text{facility}, y}$ is the net electricity generation supplied by the project to the grid, which is the difference between the electricity exported to the grid by the project ($EG_{\text{export}, y}$) and the electricity utilized by the project ($EG_{\text{import}, y}$). $EG_{\text{export}, y}$ is continuously measured by four meters (M1, M2, M3 and M4) installed in 4 loops respectively. $EG_{\text{import}, y}$ is continuously measured by the meter M5 installed at the inlet of the main transformer. All meters are bi-directional and with accuracy of 0.5s. The meter readings of $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$ were continuously measured, daily read and monthly aggregated.

The verification team had on-site checked the location of the meters against the diagram of power connection system /4/ and found them to be consistent and installed in accordance with the approved monitoring plan. All the monitoring facilities and system have been verified by CTI during on-site visit.

By checking operation log /8/ and via on-site visit, it is confirmed that both meters were working well during this monitoring period. Via the on-site interview and by checking the operation logs/8/, daily monthly reading records/12/ and electricity sales receipts confirmed by both the grid company and the project/13/, it is confirmed by the verification team that the cut-off time of electricity transaction is at 24:00 of last day of each month. Monthly reading records of five meters(M1-M5)/12/ were used to the report. The electricity sales receipts/13/ issued by the grid company and the project have been used for verifying the records of $EG_{\text{export}, y}$ and $EG_{\text{import}, y}$.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the methodology ACM0002(version 12.1.0) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

Meters	Type	SN	Accuracy class	Calibration Date	Validity	Calibration Entity
M1	DSSD331	09360003010500	0.5s	13/11/2012 06/11/2013 28/10/2014 03/07/2015 02/07/2016 21/06/2017 16/06/2018	12-11-2013 05-11-2014 27-10-2015 02-07-2016 01-07-2017 20-06-2018 15-06-2019	Shuangyashan Electricity Metering Centre
M2	DSSD331	09360003030500	0.5s	13/11/2012 06/11/2013 28/10/2014 03/07/2015 02/07/2016 21/06/2017 16/06/2018	12-11-2013 05-11-2014 27-10-2015 02-07-2016 01-07-2017 20-06-2018 15-06-2019	
M3	DSSD331	09360003130500	0.5s	13/11/2012 06/11/2013 28/10/2014 03/07/2015 02/07/2016 21/06/2017 16/06/2018	12-11-2013 05-11-2014 27-10-2015 02-07-2016 01-07-2017 20-06-2018 15-06-2019	
M4	DSSD331	09360003120500	0.5s	13/11/2012 06/11/2013 28/10/2014 03/07/2015 02/07/2016 21/06/2017 16/06/2018	12-11-2013 05-11-2014 27-10-2015 02-07-2016 01-07-2017 20-06-2018 15-06-2019	
M5	DSSD331	09360003080500	0.5s	13/11/2012 06/11/2013 28/10/2014 03/07/2015 02/07/2016 21/06/2017 16/06/2018	12-11-2013 05-11-2014 27-10-2015 02-07-2016 01-07-2017 20-06-2018 15-06-2019	

Calibration records and accreditation certificates/17//18/have been verified by the verification team. The actual accuracy of meters M1 - M5 is verified to be 0.5s, which is consistent with the requirement in the monitoring plan (accuracy degree is not less than 0.5%) contained in the registered VCS PD and in compliance with the Technical administrative code of electric energy metering /20/. The calibration frequency of meters is annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CTI confirms that:

- All meters have been installed in accordance with the monitoring plan;
- The meters' accuracy is in line with the requirement of the monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the monitoring plan.

And the calibrations are verified to be valid for the whole reporting period.

Data management and control

Shuangyashan Longyuan Wind Power Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/5/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/47/ and check records /8//12/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan contained in the VCS PD and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

By checking the UNFCCC website (<https://cdm.unfccc.int/Projects/DB/BVQI1323445785.17/view>), it is confirmed that emission reductions within the monitoring periods 19/12/2011-31/12/2012 have been issued as CERs within the 1st CDM crediting period 19/12/2011-18/12/2018.

Therefore, it is concluded the monitoring periods of the project activity is consecutive and no double counting issuance has occurred for this verified monitoring period 01/01/2013-31/12/2018 under both VCS and CDM programmes. Via the on-site interview, the project proponent confirmed the emission reductions for this verified period are only requested for VCU's issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction(CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and

energy intensive industries that emit at least 26,000 tons of CO₂e/year as per Measures for the Administration of Carbon Emission Trading/24/ including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license/19/of the project proponent, it is confirmed that the PP is qualified in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the compulsive company list under China ETS in public information/25/.

Moreover, it is confirmed by CTI the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/24/. Also, CCER mechanism has been suspended by Chinese government since 2017/26/, the project activity will not and is not allowed to registered under CCER mechanism. Thus, the assessment team concluded that the project activity not involved on other emissions trading programs and other binding limits.

The statement of no double counting/27/ submitted by PP was checked by the verification team and confirmed to be correct in terms of the regulations and database of CCER and China ETS.

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- SDG 7, Indicator 7.2 Renewable energy supplied to the power grid.
The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in NECPG. During this monitoring period, 610,513.279 MWh of net electricity generated by wind energy was delivered to the NECPG. It is verified by checking the daily and monthly reading records/12/ and via on-site inspection. This contributes to increasing the share of clean energy consumption of China's "Action Plan for Carbon Dioxide Peaking Before 2030"/40/.
- SDG 8, Indicator 8.5 Numbers of job opportunities provided by implementation of the project.
Through interviews with project proponents, employees and local villagers, verification of employee rosters, per capita wage statements provided by PP, partial labour contracts and statistics published by the national statistics data on disposable income per capita from 2013-2018/28/-/30/, it was confirmed that project activities provided 13 permanent employment opportunities for local residents and provides higher salary from 128,091 RMB/year in 2013 to 210,622 RMB/year in 2018 above the national average line level of 26,467 RMB/year in 2013 and 39,251 RMB/year in 2018. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable. This also contributes to one of the China's actions for promoting sustainable developing, "Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment" ;
- SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed.

As verified by on-site inspection, and via checking the emission reduction calculation sheet/2/ and the daily and monthly reading records/12/, 627,605 tCO₂e GHG emission reductions are generated by the project activity during this monitoring period. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas"/41/.

Meanwhile, by checking the approved VCS and CDM monitoring reports for the previous monitoring, it is confirmed the current project contributions with all impacts do not include in previously approved Sustainable Development Contribution Reports, all impacts during the previous monitoring period should not be calculated in the cumulative impact. Thus, the cumulative impacts in project lifetime are same as the one in this monitoring period.

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the project description contained in the VCS PD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved VCS PD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and wastewater, noise, solid waste and atmosphere etc. is insignificant, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the staff roster/28/, the Explanation of the employment offered by the project /29/ and the annual average salary provided by the project/30/, that the project has contributed to the local employment and improved income levels by providing job opportunities and offering permanent employees wages well above urban disposable income per capita. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training/7/ organised by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them. As verified in Section 4.1 above, the project activity contributes to achieve the China's stated sustainable development priorities in respects to increasing labor force participation rate, increasing the share

of clean energy consumption and strengthening resistance capacity to climate risks/41/. Therefore, the implementation of the project activity is beneficial to local social economic environment.

4.2.2 Local Stakeholder Consultation

The stakeholder symposium was held in Luobei County 10/05/2009. Totally 10 stakeholder representatives participated the symposium. Questionnaires were distributed according to the principle of both representation and randomness in order to reflect the public opinions and comments in a fair and real manner. Totally 40 questionnaires returned out of 40 with 100% response rate.

By checking the the summary of comments/14/ and the validation report for VCS registered PD/32/, it is confirmed that the stakeholder meeting showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, the project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in June 2014, July 2016 and May 2018 during this monitoring period. Via checking the results of the questionnaire/15/, it is confirmed by the verification team that no comments were received. Through interviewing with staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments and issues from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking VCS PD.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity supplied to the grid by the project during year y ($EG_{\text{facility}, y}$) by the validated ex-ante fixed grid emission factor ($EF_{\text{grid}, \text{CM}, y}$).

$$BE_y = EG_{\text{facility}, y} \times EF_{\text{grid}, \text{CM}, y}$$

Combined margin CO₂ emission factor for grid connected power generation ($EF_{\text{grid}, \text{CM}, y}$)

$EF_{\text{grid}, \text{CM}, y}$ is the grid emission factor of the which has been verified ex-ante in the validation stage in the VCS PD as 1.0280 tCO₂e/MWh.

The net electricity supplied by the project to the grid ($EG_{\text{facility}, y}$)

As per the Section 4.1 above, $EG_{\text{facility}, y}$ is calculated by $EG_{\text{export}, y}$ subtracting $EG_{\text{import}, y}$.

The verification team has checked the monthly reading records, electricity receipts, and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The reading records and $EG_{\text{facility}, y}$ calculation are summarized in Table 1 as follow:

Table 1 The net electricity supplied to the grid by the Project (EG_{facility, y})

Period	EG _{export,y}			EG _{import,y}			EG _{facility,y} (MWh)
	Sum of Monthly Recordings of M1-M4 (MWh)	Sales Receipt (MWh)	Conservative Quantity of the electricity exported to the grid (MWh)	Monthly Recording from M5 (MWh)	Purchase Receipt (MWh)	Conservative Quantity of the electricity imported from the grid (MWh)	
	A	B	C=min(A,B)	D	E	F=max(D,E)	
01/01/2013-31/01/2013	11,601.934	11,601.934	11,601.934	49.638	49.638	49.638	11,552.296
01/02/2013-28/02/2013	11,029.041	11,029.041	11,029.041	0.000	0.000	0.000	11,029.041
01/03/2013-31/03/2013	15,389.191	15,389.191	15,389.191	0.000	0.000	0.000	15,389.191
01/04/2013-30/04/2013	9,533.908	9,533.908	9,533.908	33.000	33.000	33.000	9,500.908
01/05/2013-31/05/2013	6,431.434	6,431.434	6,431.434	22.176	22.176	22.176	6,409.258
01/06/2013-30/06/2013	6,628.514	6,628.514	6,628.514	37.488	37.488	37.488	6,591.026
01/07/2013-31/07/2013	5,882.152	5,882.152	5,882.152	47.784	47.784	47.784	5,834.368
01/08/2013-31/08/2013	4,702.778	4,702.778	4,702.778	63.624	63.624	63.624	4,639.154
01/09/2013-30/09/2013	11,467.502	11,467.502	11,467.502	74.712	74.712	74.712	11,392.790
01/10/2013-31/10/2013	11,075.745	11,075.745	11,075.745	0.000	0.000	0.000	11,075.745
01/11/2013-30/11/2013	11,744.285	11,744.285	11,744.285	82.896	82.896	82.896	11,661.389
01/12/2013-31/12/2013	8,713.694	8,713.694	8,713.694	0.000	0.000	0.000	8,713.694
Subtotal	114,200.178	114,200.178	114,200.178	411.318	411.318	411.318	113,788.860
01/01/2014-31/01/2014	9,262.576	9,262.576	9,262.576	44.616	44.616	44.616	9,217.960
01/02/2014-28/02/2014	5,499.397	5,499.397	5,499.397	65.736	65.736	65.736	5,433.661
01/03/2014-31/03/2014	7,486.252	7,486.252	7,486.252	40.939	40.939	40.939	7,445.313

01/04/2014-30/04/2014	7,488.936	7,488.936	7,488.936	34.344	34.344	34.344	7,454.592
01/05/2014-31/05/2014	9,693.706	9,693.706	9,693.706	19.811	19.811	19.811	9,673.895
01/06/2014-30/06/2014	5,112.785	5,112.785	5,112.785	35.683	35.683	35.683	5,077.102
01/07/2014-31/07/2014	7,881.740	7,881.740	7,881.740	50.160	50.160	50.160	7,831.580
01/08/2014-31/08/2014	3,437.646	3,437.646	3,437.646	31.680	31.680	31.680	3,405.966
01/09/2014-30/09/2014	6,585.214	6,585.214	6,585.214	38.280	38.280	38.280	6,546.934
01/10/2014-31/10/2014	13,055.977	13,055.977	13,055.977	22.440	22.440	22.440	13,033.537
01/11/2014-30/11/2014	11,231.727	11,231.727	11,231.727	80.521	80.521	80.521	11,151.206
01/12/2014-31/12/2014	8,616.826	8,616.826	8,616.826	0.000	0.000	0.000	8,616.826
Subtotal	95,352.782	95,352.782	95,352.782	464.210	464.210	464.210	94,888.572
01/01/2015-31/01/2015	10,326.916	10,326.916	10,326.916	35.640	35.640	35.640	10,291.276
01/02/2015-28/02/2015	3,730.495	3,730.495	3,730.495	35.640	35.640	35.640	3,694.855
01/03/2015-31/03/2015	4,452.577	4,452.577	4,452.577	130.681	130.681	130.681	4,321.896
01/04/2015-30/04/2015	11,570.093	11,570.093	11,570.093	40.920	40.920	40.920	11,529.173
01/05/2015-31/05/2015	8,822.057	8,822.057	8,822.057	30.360	30.360	30.360	8,791.697
01/06/2015-30/06/2015	6,863.942	6,863.942	6,863.942	33.000	33.000	33.000	6,830.942
01/07/2015-31/07/2015	5,508.595	5,508.595	5,508.595	44.880	44.880	44.880	5,463.715
01/08/2015-31/08/2015	4,519.103	4,519.103	4,519.103	66.000	66.000	66.000	4,453.103
01/09/2015-30/09/2015	3,817.300	3,817.300	3,817.300	42.240	42.240	42.240	3,775.060
01/10/2015-31/10/2015	6,882.375	6,882.375	6,882.375	19.800	19.800	19.800	6,862.575
01/11/2015-30/11/2015	4,714.757	4,714.757	4,714.757	33.003	33.003	33.003	4,681.754
01/12/2015-31/12/2015	9,916.383	9,916.383	9,916.383	46.202	46.202	46.202	9,870.181
Subtotal	81,124.593	81,124.593	81,124.593	558.366	558.366	558.366	80,566.227
01/01/2016-31/01/2016	11,378.992	11,378.992	11,378.992	47.522	47.522	47.522	11,331.470

01/02/2016-29/02/2016	3,911.989	3,911.989	3,911.989	31.684	31.684	31.684	3,880.305
01/03/2016-31/03/2016	4,661.811	4,661.811	4,661.811	47.522	47.522	47.522	4,614.289
01/04/2016-30/04/2016	9,466.403	9,466.403	9,466.403	18.481	18.481	18.481	9,447.922
01/05/2016-31/05/2016	13,364.900	13,364.900	13,364.900	15.841	15.841	15.841	13,349.059
01/06/2016-30/06/2016	3,478.568	3,478.568	3,478.568	56.767	56.767	56.767	3,421.801
01/07/2016-31/07/2016	4,304.965	4,304.965	4,304.965	0.000	0.000	0.000	4,304.965
01/08/2016-31/08/2016	6,785.902	6,785.902	6,785.902	68.653	68.653	68.653	6,717.249
01/09/2016-30/09/2016	8,495.996	8,495.996	8,495.996	39.605	39.605	39.605	8,456.391
01/10/2016-31/10/2016	14,457.612	14,457.612	14,457.612	11.880	11.880	11.880	14,445.732
01/11/2016-30/11/2016	8,769.675	8,769.675	8,769.675	0.000	0.000	0.000	8,769.675
01/12/2016-31/12/2016	5,977.877	5,977.877	5,977.877	46.205	46.205	46.205	5,931.672
Subtotal	95,054.690	95,054.690	95,054.690	384.160	384.160	384.160	94,670.530
01/01/2017-31/01/2017	6,577.288	6,577.288	6,577.288	21.121	21.121	21.121	6,556.167
01/02/2017-28/02/2017	6,121.437	6,121.437	6,121.437	26.401	26.401	26.401	6,095.036
01/03/2017-31/03/2017	4,132.075	4,132.075	4,132.075	46.207	46.207	46.207	4,085.868
01/04/2017-30/04/2017	12,024.665	12,024.665	12,024.665	30.363	30.363	30.363	11,994.302
01/05/2017-31/05/2017	14,444.109	14,444.109	14,444.109	18.499	18.499	18.499	14,425.610
01/06/2017-30/06/2017	6,765.948	6,765.948	6,765.948	42.246	42.246	42.246	6,723.702
01/07/2017-31/07/2017	5,899.638	5,899.638	5,899.638	56.777	56.777	56.777	5,842.861
01/08/2017-31/08/2017	8,360.670	8,360.670	8,360.670	59.412	59.412	59.412	8,301.258
01/09/2017-30/09/2017	9,009.993	9,009.993	9,009.993	31.685	31.685	31.685	8,978.308
01/10/2017-31/10/2017	12,378.854	12,378.854	12,378.854	26.402	26.402	26.402	12,352.452
01/11/2017-30/11/2017	11,454.595	11,454.595	11,454.595	31.585	31.585	31.585	11,423.010
01/12/2017-31/12/2017	7,024.077	7,024.077	7,024.077	52.253	52.253	52.253	6,971.824

Subtotal	104,193.349	104,193.349	104,193.349	442.951	442.951	442.951	103,750.398
01/01/2018-31/01/2018	13,712.357	13,712.357	13,712.357	22.993	22.993	22.993	13,689.364
01/02/2018-28/02/2018	7,412.939	7,412.939	7,412.939	31.681	31.681	31.681	7,381.258
01/03/2018-31/03/2018	13,080.583	13,080.583	13,080.583	14.521	14.521	14.521	13,066.062
01/04/2018-30/04/2018	10,629.741	10,629.741	10,629.741	59.406	59.406	59.406	10,570.335
01/05/2018-31/05/2018	7,720.481	7,720.481	7,720.481	27.724	27.724	27.724	7,692.757
01/06/2018-30/06/2018	7,241.044	7,241.044	7,241.044	33.007	33.007	33.007	7,208.037
01/07/2018-31/07/2018	6,507.290	6,507.290	6,507.290	35.649	35.649	35.649	6,471.641
01/08/2018-31/08/2018	4,090.419	4,090.419	4,090.419	76.647	76.647	76.647	4,013.772
01/09/2018-30/09/2018	9,750.385	9,750.385	9,750.385	44.984	44.984	44.984	9,705.401
01/10/2018-31/10/2018	13,240.657	13,240.657	13,240.657	42.358	42.358	42.358	13,198.299
01/11/2018-30/11/2018	12,681.646	12,681.646	12,681.646	19.945	19.945	19.945	12,661.701
01/12/2018-31/12/2018	17,202.153	17,202.153	17,202.153	12.088	12.088	12.088	17,190.065
Subtotal	123,269.695	123,269.695	123,269.695	421.003	421.003	421.003	122,848.692
Total	613,195.287	613,195.287	613,195.287	2,682.008	2,682.008	2,682.008	610,513.279

The corresponding baseline emission reductions during this monitoring period is calculated as:

Year	EG _{facility, y} (MWh)	EF _{grid, CM, y} (tCO ₂ /MWh)	BE _y (tCO ₂)
01/01/2013-31/12/2013	113,788,860	1.0280	116,974
01/01/2014-31/12/2014	94,888.572	1.0280	97,545
01/01/2015-31/12/2015	80,566.227	1.0280	82,822
01/01/2016-31/12/2016	94,670.530	1.0280	97,321
01/01/2017-31/12/2017	103,750.398	1.0280	106,655
01/01/2018-31/12/2018	122,848.692	1.0280	126,288
Total	610,513.279	-	627,605

Project emissions

As statement in the registered VCS PD, for the wind power activities, the project emissions from the project is zero. Hence, PE_y during the monitoring period from the project activity is zero.

Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 12.1.0).

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	GHG emission reductions or removals (tCO ₂ e)
01/01/2013-31/12/2013	116,974
01/01/2014-31/12/2014	97,545
01/01/2015-31/12/2015	82,822
01/01/2016-31/12/2016	97,321
01/01/2017-31/12/2017	106,655
01/01/2018-31/12/2018	126,288
Total ERs claimed (in 2,191 days)	627,605

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates in registered CDM-PDD

The emission reductions claimed are 627,605 tCO₂e in this monitoring period (i.e. 2,191 days). Compared with expected emission reductions 750,523 tCO₂e (calculated as 125,030tCO₂/365d*2,191d), the reported emission reductions in this monitoring period are 16.38% less than the expected in the registered VCS PD. Considering the natural fluctuate of the wind resource, actual generator operation conditions and limitation of the local grid absorption capacity, CTI is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate.

Therefore, CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

The assessment of each parameter related to calculate the GHG emission reductions is justified in Sections 4.1 and 4.4 above. Via the on-site inspection and desk review, the verification team verified the supporting evidence provided by the project proponent as listed in Appendix 3 used to determine the GHG emission reductions

For each reported data, the evidence is provided and verified as sufficient and quality is appropriate. Also, the cross-checks have been performed on the reported data with different source of evidence. The information flow from data generation and aggregation, to recording, calculation and final transposition into the monitoring report has been assessed by VVB in Section 4.4. Moreover, the calibration is verified as stated in Section 4.1 via on-site inspection with evidences to confirm the meters' location, accuracy and calibration frequency are in line with requirement of the monitoring plan.

Therefore, it is concluded that the evidence provided are verified as sufficient and quality is appropriate and thus the evidence can be used to determine the GHG emission reductions and removals for this monitoring period.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions that have been reported for the project activity “Heilongjiang Bianfushan Wind Power Project” in China (VCS Project ID: 1049) for the period 01/01/2013-31/12/2018.

The verification is based on the baseline and monitoring methodology ACM0002 (version 12.1.0), the VCS PD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification approach was based on the requirements as defined under the applicable VCS Standard Version 4.3 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered VCS PD ;
- the monitoring plan is as per the applied methodology;
- the monitoring complies with the monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard Version 4.3 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

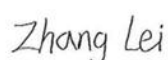
In CTI's opinion the GHG emissions reductions of the "Heilongjiang Bianfushan Wind Power Project" for the period 01/01/2013-31/12/2018 are fairly stated in the monitoring report (version 02 dated 25/07/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 12.1.0) and the monitoring plan contained in the registered VCS PD.

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Reporting period: 01/01/2013-31/12/2018

Verified GHG emission reductions and removals in the above verification period:

<i>Period</i>	Baseline emissions or removals (tCO₂e)	Project emissions or removals (tCO₂e)	Leakage emissions (tCO₂e)	Net GHG emission reductions or removals (tCO₂e)
01/01/2013-31/12/2013	116,974	0	0	116,974
01/01/2014-31/12/2014	97,545	0	0	97,545
01/01/2015-31/12/2015	82,822	0	0	82,822
01/01/2016-31/12/2016	97,321	0	0	97,321
01/01/2017-31/12/2017	106,655	0	0	106,655
01/01/2018-31/12/2018	126,288	0	0	126,288
Total	627,605	0	0	627,605



Mr.Zhang Lei

Technical Leader

21-Aug-2022



Ms. Lin Shunrong

Team Reviewer

21-Aug-2022

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCER	Chinese Certified Emission Reduction
CER	Certified Emission Reduction(s)
CL	Clarification request
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
CTI	Shenzhen CTI International Certification Co., Ltd
DOE	Designated Operational Entity
EF	Emission Factor
ER	Emission Reduction
EU ETS	EU Emissions Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas(es)
IREC	International Renewable Energy Certificates
MP	Monitoring Period
MR	Monitoring Report
NECPG	Northeast China Power Grid
PD	Project Description
PP	Project Proponent
SN	Serial Number
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: VER/VCU Monitoring Report for Heilongjiang Bianfushan Wind Power Project, V01 dated 18/07/2022, version 02 dated 25/07/2022
/2/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: Emission reduction calculation spreadsheet for Heilongjiang Bianfushan Wind Power Project, V01 dated 18/07/2022
/3/	Shuangyashan Longyuan Wind Power Co., Ltd. and State Grid Heilongjiang Electric Power Ltd.: Power Purchase Agreement for State Grid Heilongjiang Electric Power Ltd.
/4/	Shuangyashan Longyuan Wind Power Co., Ltd. : Diagram of power connection system
/5/	Shuangyashan Longyuan Wind Power Co., Ltd.: Map and layout of the project site
/6/	Shuangyashan Longyuan Wind Power Co., Ltd.: VER monitoring manual and management procedure.
/7/	Shuangyashan Longyuan Wind Power Co., Ltd.: Records of training for on-site staff.
/8/	Shuangyashan Longyuan Wind Power Co., Ltd.: Operation logs for project activity, during the period of 25/12/2009-24/12/2019.
/9/	Shuangyashan Longyuan Wind Power Co., Ltd.: Application Report for Commencement of Work
/10/	Project construction-start application and permission, dated 04/06/2009.
/11/	Beijing Zhubiao Construction Engineering Consulting Co., Ltd. and Beijing Zhubiao Certified Public Accountants Co., Ltd.: Verification report of project construction completion of Heilongjiang Bianfushan Wind Power Project, dated 31/08/2010.
/12/	Shuangyashan Longyuan Wind Power Co., Ltd.: Monthly reading records of Heilongjiang Bianfushan Wind Power Project from 01/01/2013-31/12/2018
/13/	State Grid Heilongjiang Electric Power Ltd.: Electricity sales receipts for electricity exported to and imported from the grid by the project activity for 01/01/2013-31/12/2018

/14/	Meeting memo of the stakeholder consultation and the participants' signature form, 10/05/2009
/15/	Questionnaires for local stakeholders
/16/	Shuangyashan Longyuan Wind Power Co., Ltd.: Registered CDM PDD (version 03 dated 01/11/2011)
/17/	Heilongjiang Provincial Market Supervision Administration: Accreditation certificate of Shuangyashan Electricity Metering Centre, issued on 22/06/2012, valid up to 21/06/2016; issued on 22/06/2016, valid up to 21/06/2020
/18/	Shuangyashan Electricity Metering Centre: Calibration certificates for meters M1, M2, M3, M4 and M5 covering this monitoring period
/19/	Business license of Shuangyashan Longyuan Wind Power Co., Ltd.
/20/	State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T448)
/21/	UNFCCC, http://cdm.unfccc.int
/22/	Gold Standard, http://www.goldstandard.org/
/23/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/24/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/25/	Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/26/	Announcement of the National Development and Reform Commission, Code 2 in 2017, issued by National Development and Reform Commission, dated 03/14/2017
/27/	Shuangyashan Longyuan Wind Power Co., Ltd.: Statement of no double counting
/28/	Shuangyashan Longyuan Wind Power Co., Ltd.: Staff roster
/29/	Shuangyashan Longyuan Wind Power Co., Ltd.: Explanatory material on the number of people employed and the average salary provided by the project.
/30/	Statistical Bulletin on National Economic and Social Development by Year: http://www.stats.gov.cn/search/
/31/	DNV Climate Change Services AS: VCS Validation Report (included in verification Report No. 2013-9087, dated 12/04/2013)
/32/	VER/VCU Monitoring Report for Heilongjiang Bianfushan Wind Power Project of previous monitoring period 25/12/2009-18/12/2011

/33/	DNV Climate Change Services AS: VCS Verification Report for Heilongjiang Bianfushan Wind Power Project of previous monitoring period 25/12/2009-18/12/2011
/34/	Bureau Veritas Certification Holding SAS (BVCH): CDM Validation Report, CHINA-Val/0232/2009, revision 01 dated 07/11/2011
/35/	CDM/CER Monitoring Reports for Heilongjiang Bianfushan Wind Power Project of previous CDM monitoring periods 19/12/2011-31/12/2012
/36/	CDM Verification Reports for Heilongjiang Bianfushan Wind Power Project of previous monitoring period 19/12/2011-31/12/2012
/37/	General Electric (Shenyang) Co., Ltd.: Nameplates of the equipment
/38/	Feasibility Study Report of Heilongjiang Bianfushan Wind Power Project
/39/	Co-signed by the General Electric (Shenyang) Co., Ltd. and Shuangyashan Longyuan Wind Power Co., Ltd.: Technical agreement of wind turbines purchase contract for the project activity Heilongjiang Bianfushan Wind Power Project, dated 14/05/2009
/40/	Action Plan for Carbon Dioxide Peaking Before 2030 published by the State Council on 24/10/2021
/41/	China's National Programme to Address Climate Change published by the State Council on 28/03/2008

Methodologies, tools and other guidance

/42/	Verified Carbon Standard: VCS Standard, version 4.3
/43/	Verified Carbon Standard: VCS Program Guide, version 4.2
/44/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/45/	Verified Carbon Standard: Registration and Issuance Process, version 4.1
/46/	UNFCCC EB: Approved methodology, ACM0002 , version 12.1.0

Persons interviewed

/47/	Mr. Wang Zhewei, Station Manager, Shuangyashan Longyuan Wind Power Co., Ltd. Mr. Zhao Guochao, Operator, Shuangyashan Longyuan Wind Power Co., Ltd. Mr. Fu Tao, Operator, Shuangyashan Longyuan Wind Power Co., Ltd. Ms. Yang Huifang, Local resident, Local Village Mr. Li Jianhong, Local resident, Local Village
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	Mr. Zhao Wei, Staff, Local Environment Protection Bureau Ms. Ma Qian, Project Manager, Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
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APPENDIX C: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Lei ZHANG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.1: Thermal energy generation
	TA 1.2: Energy generation from renewable energy sources
SS 4: Manufacturing industries	TA 4.1: Cement and lime production
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

Ms. Wenting ZHANG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

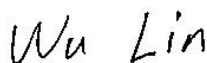
Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	-	-	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu LIN



Technical Competent Manager

Shenzhen, 01/03/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX D: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	It is requested to clarify the manufacturer of the turbine-generator.	The manufacturer of the turbine-generator had been updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR. Therefore, CL 01 is closed.

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA